CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 2 OF 10

# Data Analyst Interview Preparation: SQL, Excel, Power BI, Statistics and Case Questions

Prepare for analyst interviews with the question types that appear most often, model answers and practice links: SQL coding rounds (joins, aggregation, window functions), Excel and Power BI questions, statistics and A/B testing, metric and product-sense cases, and behavioural questions with the STAR method.

Career track · 2026 · Intermediate · 120 min

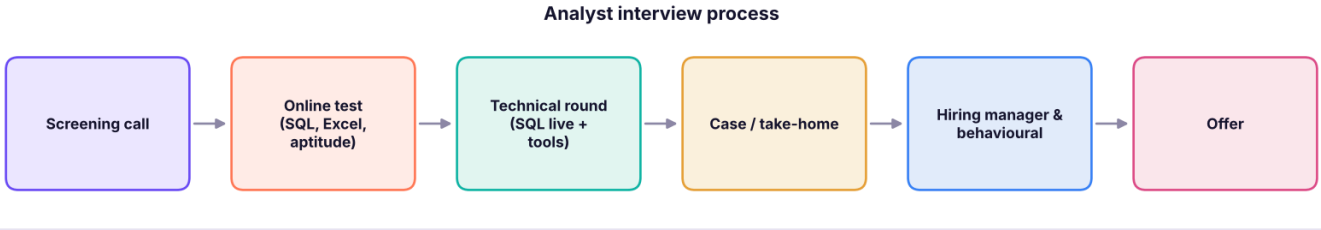
**By the end of this lesson you can**

- Solve common SQL interview patterns quickly and explain your approach
- Answer core Excel, Power BI and DAX questions
- Explain key statistics and A/B testing concepts clearly
- Structure metric, root-cause and product-sense cases
- Tell project stories with STAR

**Datasets used**

`superstore_sales.csv`

## The interview loop




A typical analyst hiring process.

# SQL patterns you must know

Python

```
import duckdb, pandas as pd
sales = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
print(duckdb.sql("""
-- Top 2 products by sales in each region (window function)
WITH r AS (
  SELECT region, product_name, SUM(sales) AS sales,
         RANK() OVER (PARTITION BY region ORDER BY SUM(sales) DESC) AS rnk
  FROM sales GROUP BY region, product_name)
SELECT region, product_name, ROUND(sales) AS sales FROM r WHERE rnk ⩽ 2 ORDER BY region,
rnk;
""").df())
```

OUTPUT

| | region | product_name | sales |
|---|---|---|---|
| 0 | East | Apex Copier 347 | 4191180.0 |
| 1 | East | Zenith Copier 719 | 3323372.0 |
| 2 | North | Orbit Copier 746 | 3790940.0 |
| 3 | North | Zenith Machine 401 | 3566453.0 |
| 4 | South | Orbit Copier 746 | 5405517.0 |
| 5 | South | Vertex Copier 262 | 4258600.0 |
| 6 | West | Zenith Copier 719 | 4087068.0 |
| 7 | West | Zenith Machine 401 | 4023468.0 |

Python

```
import duckdb, pandas as pd
sales = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
print(duckdb.sql("""
-- Month-over-month growth with LAG
WITH m AS (SELECT DATE_TRUNC('month', order_date) AS month, SUM(sales) AS sales FROM sales
GROUP BY 1)
SELECT month, ROUND(sales) AS sales, ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY
month)) / LAG(sales) OVER (ORDER BY month), 1) AS mom_pct
FROM m ORDER BY month DESC LIMIT 4;
""").df())
```

OUTPUT

| | month | sales | mom_pct |
|---|---|---|---|
| 0 | 2025-12-01 | 7652267.0 | -25.6 |
| 1 | 2025-11-01 | 10284516.0 | 67.5 |
| 2 | 2025-10-01 | 6140290.0 | -27.5 |
| 3 | 2025-09-01 | 8468060.0 | 45.9 |

| Pattern | Typical question |
|---|---|
| GROUP BY + HAVING | Customers with more than 5 orders |
| Joins (inner, left, anti) | Customers who never ordered |
| Window: ROW_NUMBER/RANK | Top N per group, de-duplicate latest record |
| Window: LAG/LEAD, running SUM | Growth, retention, cumulative totals |
| CASE WHEN | Buckets, conditional counts, pivots |
| Date functions | Monthly cohorts, days between events |
| Self join / recursive CTE | Manager hierarchy, consecutive days |

Approach out loud: clarify the table grain → write the query step by step (CTEs) → check edge cases (NULLs, ties, duplicates) → validate with a small count.

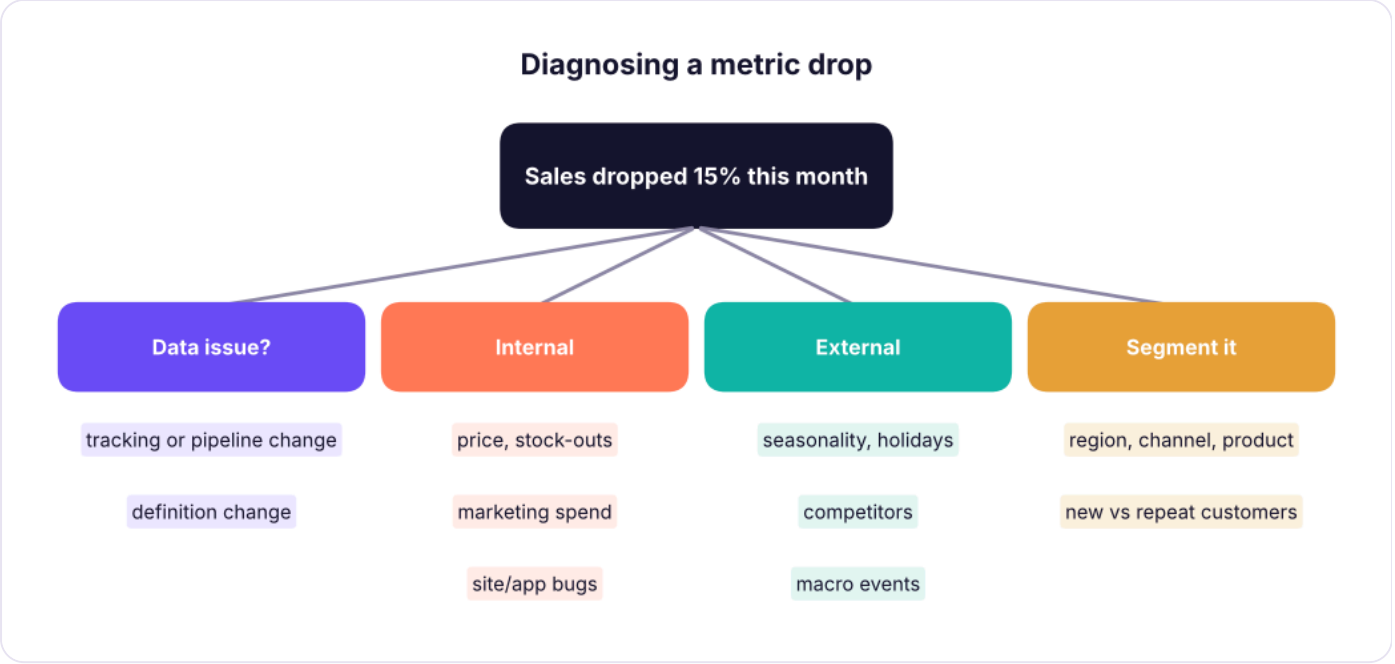
## Excel and Power BI questions

| Question | Short answer |
|---|---|
| VLOOKUP vs XLOOKUP vs INDEX-MATCH? | XLOOKUP looks left/right, has defaults and exact match by default; INDEX-MATCH works in older versions; VLOOKUP breaks when columns move |
| Pivot table vs Power Pivot? | Power Pivot uses a data model with relationships and DAX measures across tables |
| Calculated column vs measure (DAX)? | Columns are computed row by row and stored; measures are evaluated in filter context at query time |
| What does CALCULATE do? | Evaluates an expression in a modified filter context |
| Star schema? | Fact table of events joined to dimension tables; simpler, faster models |
| Import vs DirectQuery? | Import copies data (fast, scheduled refresh); DirectQuery queries the source live (fresh, slower) |
| Row-level security? | Roles with DAX filters so users see only their rows |

## Statistics and experiments

Be ready to explain: mean vs median, standard deviation, distributions, correlation vs causation, confidence intervals, p-values, Type I/II errors, sample size and A/B test pitfalls (peeking, SRM, novelty). See ML lessons 1–6 for model answers.

## Case questions: a framework




Root-cause framework for "metric X dropped".

1. Clarify the metric and time frame. 2. Check data quality first. 3. Segment (where is the drop?). 4. Form hypotheses (internal vs external). 5. Validate with data. 6. Recommend action and how you would measure it.

Product-sense metrics: define a north-star metric, input metrics and guardrails (e.g. for a food-delivery app: weekly ordering customers; conversion, delivery time, order value; cancellations, complaints).

## Behavioural: STAR

**Situation** (context) → **Task** (your goal) → **Action** (what *you* did, with tools) → **Result** (numbers, learning). Prepare 6 stories: a project you are proud of, a mistake you fixed, a conflict, a tight deadline, influencing without authority, learning a new tool fast.

## Practice resources

W3Schools SQL quiz and exercises, LeetCode SQL 50, HackerRank SQL, StrataScratch, DataLemur, Kaggle Learn; mock interviews with peers; time yourself (15 minutes per medium SQL question).

KEY TAKEAWAYS

- Master SQL patterns: aggregation, joins, window functions, CTEs, dates; think aloud and check edge cases.
- Know Excel lookups/pivots and Power BI modelling, DAX, CALCULATE, star schema and RLS.
- Explain statistics and A/B testing simply.
- Use a structured framework for metric and product cases.
- Prepare STAR stories with quantified results.

## Quick quiz

4 questions · tap an answer to check it

**1. Which SQL feature finds the top 3 products in each category?**

**A** A window function such as RANK() OVER (PARTITION BY category ORDER BY sales DESC)

**B** GROUP BY only

**C** DISTINCT

**D** UNION

**Answer: A.** Then filter rank ≤ 3.

**2. Sales dropped 15%. What is the first check?**

**A** Whether the data or tracking changed

**B** Fire the marketing team

**C** Build an ML model

**D** Ignore it

**Answer: A.** Rule out data issues first.

**3.** What is a DAX measure?

- **A** An expression evaluated at query time in the current filter context
- **B** A stored column
- **C** A table relationship
- **D** A Power Query step

**Answer: A.** Columns are stored row by row.

**4.** What does STAR stand for?

- **A** Situation, Task, Action, Result
- **B** Skills, Tools, Analysis, Reporting
- **C** Start, Test, Assess, Repeat
- **D** None

**Answer: A.** Structure for behavioural answers.

EXERCISE **Practice**

(1) Write SQL for: customers who ordered in 2024 but not in 2025; the second-highest order per customer; 7-day moving average of sales. (2) Prepare two STAR stories. (3) Do a 30-minute mock case: "Delivery complaints doubled last week."

**Show solution (worked answer)**

Use NOT EXISTS / LEFT JOIN ... IS NULL for the anti-join, DENSE_RANK() = 2 for the second highest, and AVG(...) OVER (ORDER BY day ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) for the moving average.

## Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.